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United States Patent Application Publication

20250275613

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Ness; Kelly

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### Automatic Stand Utility Bag

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#### Abstract

The present invention discloses an Automatic Stand Utility Bag featuring an integrated, automatically extensible stand for improved stability and accessibility. The bag comprises a tubular structure with a fabric sleeve and rigid materials for enhanced durability and stiffness. The stand includes pivotally mounted legs with upper, intermediate, and lower segments that extend outward upon setting the bag down and then tilting the bag from a vertical to a leaning position. An actuator rod facilitates automatic deployment, ensuring the bag remains stable on various surfaces. Reflective material enhances visibility, while accessory pockets provide organization for smaller items. The innovative design eliminates manual setup and combines convenience, compactness, and safety, making it an ideal choice for users in construction, workshops, or outdoor environments. This utility bag significantly improves safety, efficiency, and ergonomic utility for professionals and hobbyists alike.

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**Inventors:** Ness; Kelly (St. Louis Park, MN)

**Applicant:** Ness; Kelly (St. Louis Park, MN)

**Family ID:** 96881340

**Appl. No.:** 19/036362

**Filed:** January 24, 2025

#### Related U.S. Application Data

us-provisional-application US 63561255 20240304

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#### Publication Classification

**Int. Cl.:** A45C5/08 (20060101); A45C5/02 (20060101); A45C13/00 (20060101); A45C13/02 (20060101); A45C13/10 (20060101); A45C13/36 (20060101)

CPC     **A45C5/08** (20130101); **A45C5/02** (20130101); **A45C13/008** (20130101); **A45C13/02** (20130101); **A45C13/103** (20130101); A45C2013/026 (20130101); A45C13/36 (20130101); A45C2200/15 (20130101)

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## Background/Summary

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### BACKGROUND

#### Field of the Invention

[0002] The invention relates to the field of utility bags, specifically those used for carrying tools or accessories in professional or recreational settings. It addresses the challenges of stability, accessibility, and organization by incorporating an automatically extensible stand. The invention enhances usability by improving safety, convenience, ergonomics, and efficiency, making it suitable for diverse environments, including construction sites, workshops, and outdoor activities.

#### Description of the Related Art

[0003] Utility bags have long been indispensable tools for professionals, hobbyists, and outdoor enthusiasts, enabling them to carry and organize tools and materials conveniently. From construction workers who rely on these bags to store heavy-duty tools to electricians who need quick access to specific equipment, utility bags serve as an essential accessory in a variety of work environments. However, traditional designs often prioritize basic storage over functionality, leading to inefficiencies in use. Many utility bags fail to provide stability or easy accessibility in real-world conditions, such as uneven terrain or confined spaces, resulting in a less-than-optimal user experience. This has created a demand for a solution that goes beyond the limitations of conventional utility bags.

[0004] The importance of utility bags in improving workplace efficiency and safety cannot be overstated. For professionals working in demanding environments, every second spent searching for a tool or stabilizing a bag represents lost productivity. Moreover, the lack of stability in most utility bags leads to the need for constant repositioning, which can cause frustration and fatigue, particularly over long hours of use. Accessibility is equally critical, as bending repeatedly to retrieve tools from an unstable bag on the ground can lead to back strain and other ergonomic issues. Additionally, visibility plays a vital role in ensuring safety, especially in high-traffic or low-light environments such as construction sites or roadside projects. A utility bag that addresses these factors can significantly enhance both the efficiency and safety of its users.

[0005] Recent studies highlight the widespread challenges faced by utility bag users, underlining the need for innovation in this area. Surveys conducted among professionals in construction, repair, and maintenance industries reveal that over 70% find the lack of stability in utility bags a significant issue, particularly when working outdoors. Furthermore, nearly 65% of users reported difficulties in accessing tools quickly when using conventional designs. The global tool storage market has grown significantly in recent years, reaching an estimated \$10 billion in value, with an annual growth rate of 5%. This growth reflects the increasing demand for utility products that offer better functionality, durability, and safety features. Despite this, most existing utility bags fail to address the core issues of stability, accessibility, and ergonomic convenience.

[0006] In response to the growing demand for better tool storage solutions, various manufacturers have introduced innovations aimed at improving utility bags. These include reinforced bottoms for better durability, modular compartments for customizable storage, and wheels for improved mobility. Some designs have even incorporated lightweight materials and ergonomic handles to reduce strain during transport. However, when it comes to stability, solutions have been limited. Some bags feature detachable stands or manual support mechanisms, but these systems often require additional effort to assemble or are cumbersome to use. While these innovations address some user needs, they tend to fall short in offering an all-in-one solution that seamlessly integrates stability, accessibility, and portability.

[0007] Despite advancements in utility bag designs, several critical shortcomings persist. The most glaring issue is the lack of stability, particularly when the bag is placed on uneven surfaces or inclined planes. This problem is compounded by the absence of a reliable, easy-to-use mechanism for keeping the bag upright. Many existing solutions rely on makeshift or detachable stands, which are often inconvenient and detract from the bag's portability. Another major issue is accessibility. Traditional utility bags, which lie flat on the ground, force users to bend repeatedly to retrieve tools, leading to discomfort and inefficiency. Durability also remains a concern, as many bags are unable to withstand harsh environmental conditions, such as rain, mud, or abrasive surfaces. Additionally, safety features such as reflective materials are rarely incorporated, leaving users in low-light or high-traffic conditions vulnerable to accidents. These shortcomings underscore the need for a comprehensive solution that addresses all these pain points.

[0008] The Automatic Stand Utility Bag is a revolutionary solution designed to address the long-standing challenges faced by utility bag users. Unlike conventional designs, this bag incorporates an automatically extensible stand that deploys seamlessly when the bag is tilted, eliminating the need for manual setup. The stand is integrated into the bag's body and operates through an innovative actuator rod mechanism, which allows the legs to extend outward in a stable configuration. This automatic functionality provides unparalleled convenience, as the bag remains upright and accessible on a wide range of surfaces, including uneven or sloped terrain. Additionally, the bag is equipped with reflective materials on its upper portion to enhance visibility in low-light conditions, as well as multiple accessory pockets to improve organization and efficiency. Constructed from lightweight yet durable materials, the bag offers the perfect balance of portability and resilience, making it suitable for professionals in demanding work environments.

[0009] The Automatic Stand Utility Bag fills a critical void in the current market by offering an integrated, user-friendly solution to the problems of stability, accessibility, and portability. Unlike traditional bags that require manual adjustments or rely on separate stands, this bag features a built-in mechanism that automatically deploys the stand when needed. This not only enhances user convenience but also significantly improves workflow efficiency. The bag's compact design ensures it remains portable, even with the added functionality of the stand. Its durable materials and safety features, such as reflective material, make it ideal for outdoor and high-risk environments. By addressing these issues comprehensively, the Automatic Stand Utility Bag meets the needs of modern users who demand both practicality and innovation in their tools. It is not just a tool storage solution but a reimagined utility product that sets a new standard in the industry.

[0010] None of the previous inventions and patents, taken either singly or in combination, is seen to describe the instant invention as claimed. Hence, the inventor of the present invention proposes to resolve and surmount existent technical difficulties to eliminate the aforementioned shortcomings of prior art.

## SUMMARY

[0011] In light of the disadvantages of the prior art, the following summary is provided to facilitate an understanding of some of the innovative features unique to the present invention and is not intended to be a full description. A full appreciation of the various aspects of the invention can be gained by taking the entire specification, claims, drawings, and abstract as a whole.

[0012] The present invention seeks to improve prior techniques and provide an advanced design and methodology for bag with an extensible stand.

[0013] The invention also aims to provide a utility bag with an automatically extensible stand that enhances stability on various surfaces.

[0014] A further objective of the present invention is to simplify the deployment process by automating the stand's extension when the bag is set down.

[0015] It is also an object of the invention is to improve user ergonomics by eliminating the need for manual setup of the stand.

[0016] It is further the objective of the invention to maintain a compact and portable design by allowing the stand to retract against the bag body.

[0017] It is also the objective of the invention to offer a durable and lightweight construction suitable for heavy-duty use.

[0018] The objective of the invention is also to increase safety and visibility through reflective material applied to the upper portion of the bag.

[0019] It is also the objective of the invention to facilitate organization with accessory pockets for tools and smaller items.

[0020] It is further the objective of the invention to enhance accessibility to contents, reducing strain and increasing efficiency in retrieving items.

[0021] The invention aims to provide a durable stand with materials resistant to wear, corrosion, and environmental factors, ensuring longevity.

[0022] Thus, the objective is to support versatile applications in diverse environments, such as construction sites, workshops, and outdoor activities.

[0023] This Summary is provided merely for purposes of summarizing some example embodiments, so as to provide a basic understanding of some aspects of the subject matter described herein. Accordingly, it will be appreciated that the above-described features are merely examples and should not be construed to narrow the scope or spirit of the subject matter described herein in any way. Other features, aspects, and advantages of the subject matter described herein will become apparent from the following Detailed Description, Figures, and Claims.

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## Description

### DESCRIPTION OF THE DRAWINGS

[0024] The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments of concepts that include the claimed invention and explain various principles and advantages of those embodiments.

[0025] FIG. 1 illustrates the utility bag with its stand fully deployed as per preferred embodiments of the invention.

[0026] FIG. 2 displays the utility bag in its compact, retracted state as per preferred embodiments of the invention.

[0027] Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present invention.

[0028] The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the

description herein.

## DETAILED DESCRIPTION

[0029] Detailed descriptions of the preferred embodiment are provided herein. It is to be understood, however, that the present invention may be embodied in various forms. Therefore, specific details disclosed herein are not to be interpreted as limiting, but rather as a basis for the claims and as a representative basis for teaching one skilled in the art to employ the present invention in virtually any appropriately detailed system, structure or manner.

[0030] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well as the singular forms, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

[0031] The present invention referred to as the Automatic Stand Utility Bag, combines the portability and functionality of a standard utility bag with an integrated stand system that deploys automatically. This feature solves common issues such as lack of stability, difficult access to tools, and manual setup. The bag's ergonomic design enhances efficiency for professionals in various environments, including construction sites, workshops, and outdoor activities.

[0032] As per its proffered embodiments, the bag's main body has a tubular shape reinforced with a rigid liner running vertically along its length. This liner is enclosed within a durable fabric sleeve, providing both flexibility and resilience under different conditions. The rigid liner is made of lightweight materials such as aluminum or composite polymers to balance durability with portability. The fabric, treated for water and abrasion resistance, ensures the bag withstands daily wear and tear.

[0033] As per its further embodiments, the accessory pockets are positioned strategically on the bag's exterior for easy access to smaller items, paint cans, wood stakes like screws, nails, or hand tools. These pockets may use secure closures such as zippers, elastic bands, or Velcro to prevent items from falling out during transport or use.

[0034] As per its further embodiments, the defining feature of the utility bag is its automatically extensible stand. The stand comprises two legs, each constructed with upper, intermediate, and lower segments. These segments are pivotally connected, allowing the legs to fold into a compact, retracted position or extend outward in a stable, deployed configuration. The pivot points are designed with reinforced joints to handle frequent movement and weight stress.

[0035] As per its further embodiments, the actuator rod is the core component of the deployment mechanism. Positioned along the bottom of the bag, the rod is connected to the two legs via pivoting mounts. When the bag is set down, then tilted from its vertical position, the rod activates and pushes the legs outward, causing them to deploy. This tilting action is intuitive for users, eliminating the need for manual adjustments.

[0036] As per its further embodiments, in its deployed state, the stand provides stability on uneven surfaces by spreading the legs in a nonparallel arrangement. This configuration prevents tipping and ensures the bag remains upright, even under significant loads.

[0037] As per its further embodiments, when the bag is returned to an upright position, the actuator rod retracts the legs into their original, parallel position. The retraction process is smooth and requires minimal user effort. In the retracted state, the legs align against the bag's body, preserving its sleek profile and ensuring easy transportation.

[0038] As per its further embodiments, the utility bag incorporates reflective materials on its upper sections for enhanced visibility. These materials are strategically placed to provide 360-degree

visibility in low-light or high-traffic environments. Workers operating in dimly lit conditions, such as construction sites or roadways, benefit from increased safety.

[0039] As per its further embodiments, the reflective material is embedded into seams or applied as strips, ensuring durability and resistance to wear. Additionally, the bag may meet safety standards for use in hazardous environments.

[0040] As per its further embodiments, the bag includes multiple accessory pockets tailored for storing various tools and items. These pockets are made from reinforced fabric to carry heavy loads without tearing. Modular designs allow users to rearrange or detach the pockets as needed. The interior of the bag is spacious, accommodating larger tools or equipment.

[0041] As per its further embodiments, the utility bag is designed to function in diverse environmental conditions. The materials used in the bag and stand are resistant to corrosion, moisture, and UV damage. The legs are equipped with non-slip pads at their lower ends, ensuring stability on slippery or uneven surfaces. This adaptability makes the bag suitable for both indoor and outdoor use.

[0042] As per its further embodiments, the bag's versatility is a key advantage. It is ideal for professionals such as surveyors, construction workers, electricians, and plumbers who require easy access to tools. Outdoor enthusiasts, like gardeners, campers or hikers, will also find the bag useful due to its stability and compact design. The automatic stand enhances usability across various terrains, from workshop floors to rugged outdoor trails.

[0043] As per its further embodiments, FIG. 1 illustrates the utility bag with its stand fully deployed. The legs extend outward in a nonparallel arrangement, forming a wide base that ensures stability. This configuration is particularly advantageous on uneven terrain, preventing tipping even when the bag is heavily loaded. The actuator rod is shown in its extended position, demonstrating the mechanism that triggers deployment upon tilting the bag.

[0044] As per its further embodiments, the FIG. 2 displays the utility bag in its compact, retracted state. The legs are neatly aligned parallel to the bag's body, preserving its streamlined profile. The retraction mechanism allows the legs to fit snugly against the bag, ensuring portability. The actuator rod is shown in its original position, ready to activate when the bag is tilted again.

[0045] While a specific embodiment has been shown and described, many variations are possible. With time, additional features may be employed. The particular shape or configuration of the platform or the interior configuration may be changed to suit the system or equipment with which it is used.

[0046] Having described the invention in detail, those skilled in the art will appreciate that modifications may be made to the invention without departing from its spirit. Therefore, it is not intended that the scope of the invention be limited to the specific embodiment illustrated and described. Rather, it is intended that the scope of this invention be determined by the appended claims and their equivalents.

[0047] The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

## Claims

**1.** An Automatic Stand Utility Bag, comprising: A tubular utility bag body with rigid material for structural support; A stand mechanism attached to the bag body, including pivotally mounted first and second legs with upper, intermediate, and lower segments; An actuator rod connected to the legs, enabling movement from a retracted position to a deployed position; and, A deployment mechanism is triggered when the bag transitions from a vertical to a leaning position, providing stability on various surfaces. The utility bag of claim 1, wherein the legs are constructed from lightweight aluminum or composite material to ensure portability and durability. The utility bag of claim 1, wherein the actuator rod is spring-loaded to facilitate smooth and automatic deployment of the stand. The utility bag of claim 1, wherein the legs are designed to deploy at a predefined angle for maximum stability on uneven terrain. The utility bag of claim 1, wherein the deployment mechanism uses gravity and the weight of stored items to secure the legs in the deployed position. The utility bag of claim 1, wherein the retracted position of the stand mechanism allows the legs to align against the body of the bag for compact storage. The utility bag of claim 1, wherein the actuator rod is adjustable to modify the angle of the deployed legs as required. The utility bag of claim 1, wherein the stand mechanism is resistant to corrosion and environmental wear.

**2.** The utility bag further comprising reflective material applied to the upper portion of the bag for enhanced visibility, especially in low-light conditions. The utility bag of claim 2, wherein the reflective material is positioned in a 360-degree pattern to ensure visibility from all directions. The utility bag of claim 2, wherein the reflective material includes phosphorescent properties for added illumination. The utility bag of claim 2, wherein the reflective material is incorporated into the seams of the bag for seamless integration. The utility bag of claim 2, wherein the reflective material is designed to meet safety standards for outdoor and roadside use. The utility bag of claim 2, wherein the reflective material is water-resistant to maintain functionality in adverse weather conditions. The utility bag of claim 2, wherein the reflective material is customizable in color or pattern for branding purposes. The utility bag of claim 2, wherein the reflective material may be sewn or heat-applied for improved durability.

**3.** The utility bag further comprising accessory pockets on the outer surface of the bag for organized storage of smaller tools and items. The utility bag of claim 3, wherein the accessory pockets may be equipped with elastic closures to prevent items from falling out. The utility bag of claim 3, wherein the accessory pockets may include zippered compartments for secure storage. The utility bag of claim 3, wherein the accessory pockets may be padded to protect fragile items. The utility bag of claim 3, wherein the accessory pockets may be modular and can be detached or repositioned. The utility bag of claim 3, wherein the accessory pockets are made from waterproof material to protect the contents from moisture. The utility bag of claim 3, wherein the accessory pockets may include mesh designs for easy identification of contents. The utility bag of claim 3, wherein the accessory pockets may feature reinforced stitching for added durability under heavy use.

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